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$$\begin{aligned} & \int_{-\infty}^{-1} \frac{dx}{x^3} \\ &= \lim_{a \rightarrow -\infty} \int_a^{-1} x^{-3} dx \\ &= \lim_{a \rightarrow -\infty} \left[-\frac{1}{2x^2} \right]_a^{-1} \\ &= \lim_{a \rightarrow -\infty} \left[-\frac{1}{2} + \frac{1}{2a^2} \right] \\ &= \left[-\frac{1}{2} + 0 \right] \\ &= -\frac{1}{2} \end{aligned}$$

References